Industry 4.0

Twenty Fourth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Industry 4.0





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First Industrial Revolution (1760-1840)
Key features

- Extensive use of iron and steel
- Steam engine and later IC engines
- Automation in textile sector
- Emergence of factory system: Work from home was replaced by industrial clusters. Are we now taking 360 degree turn?
- Use of railways
- A lot of application of science to engineering
- Industry revolution spread slowly from Britain to Belgium, France

Second Industrial Revolution (1870-1914)
Key features

- Use of electrical energy
- Mass production
- Assembly lines
- Automation
- Development of special purpose machine tools
- Use of non-ferrous materials, aluminum, plastics
- Development of phones (1876), bulbs

Third Industrial Revolution (1970-1990)
Key features

- Electronics, Telecommunication, computers
- CNC machines
- Programmable logic controllers
- Robotics
- Nuclear power
- Internet (Ray Tomlinson invented e-mail in 1971. Internet came into existence in 1973. Tim Berners-Lee developed World Wide Web in 1991.)

(Many consider the invention of transistor in 1947 by John Bardeen, Walter Brattain and William B. Shockley as a revolutionary event.)

(The first electronic computer ENIAC (Electronic Numerical Integrator and Calculator) developed in 1946 had about 18000 thermionic valves and consumed about 150 kW of electrical power.)

Fourth Industrial Revolution (present)
Key features

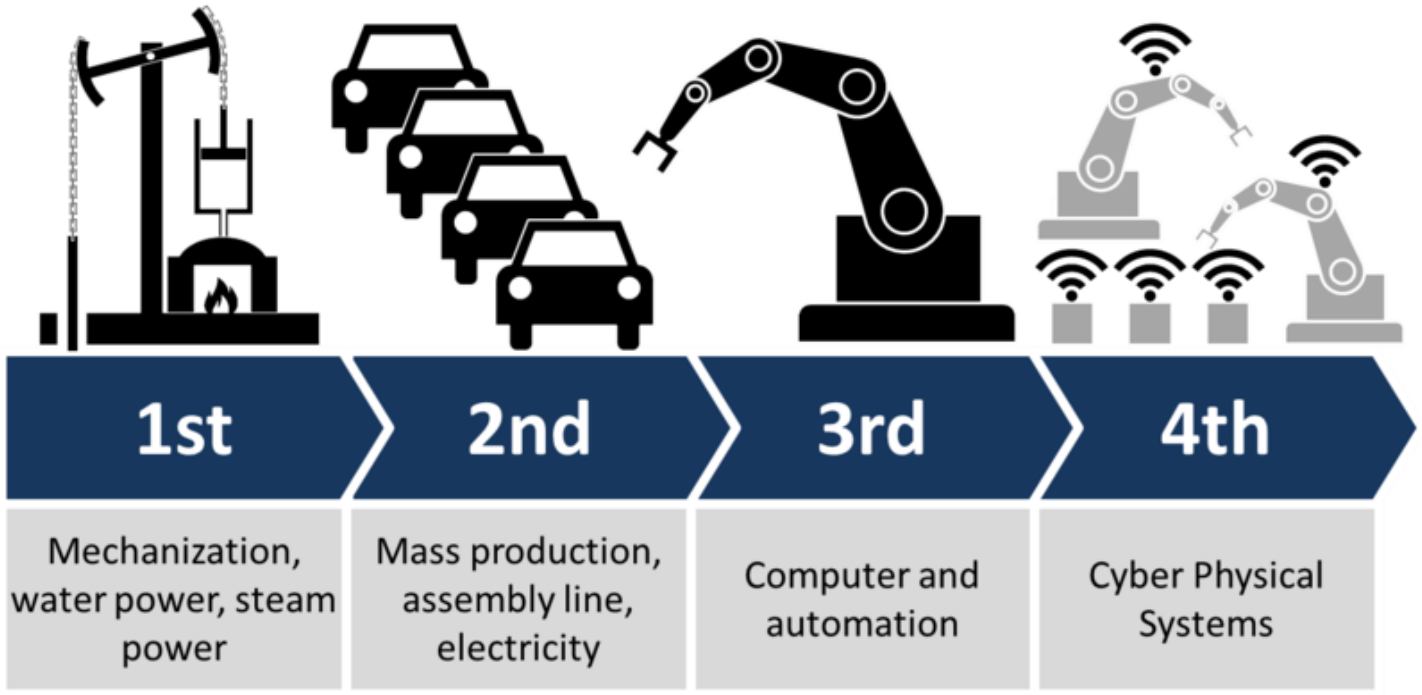
- Named before its advent
- Cyber physical systems
- Short development period
- Mass customization/personalization (Personalization is done through customer data)
- Flexibility
- Decentralization (Will factory system go away?)
- Resource efficiency
- Web 2.0
- Smart Phones

Contd.
Fourth Industrial Revolution
Key features

- Additive manufacturing, 3D printers
- Autonomous manufacturing cells
- Digitization and networking (digital manufacturing)
- Digital twins (virtual replica of physical devices)
- Miniaturization
- Smart factory
- Importance of IT

Industry 4.0

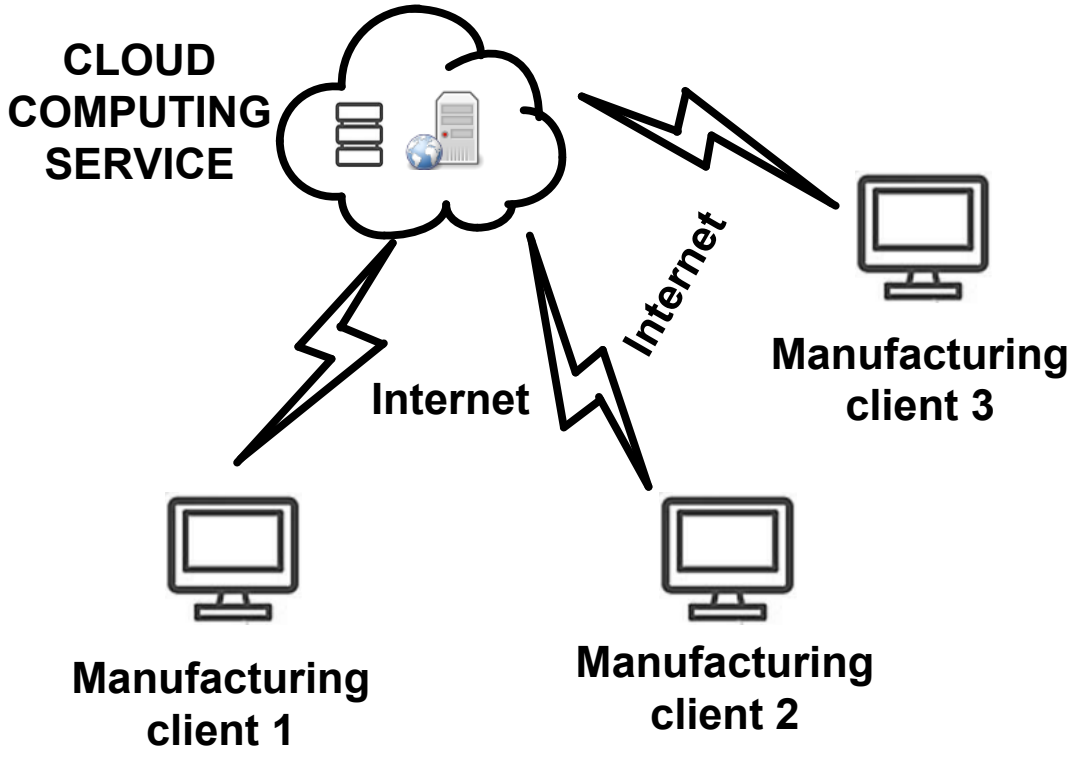
- Main ideas were published by Kagermann in 2011.
- German National Academy of Science and Engineering



Credit: "*Christoph Roser at AllAboutLean.com*

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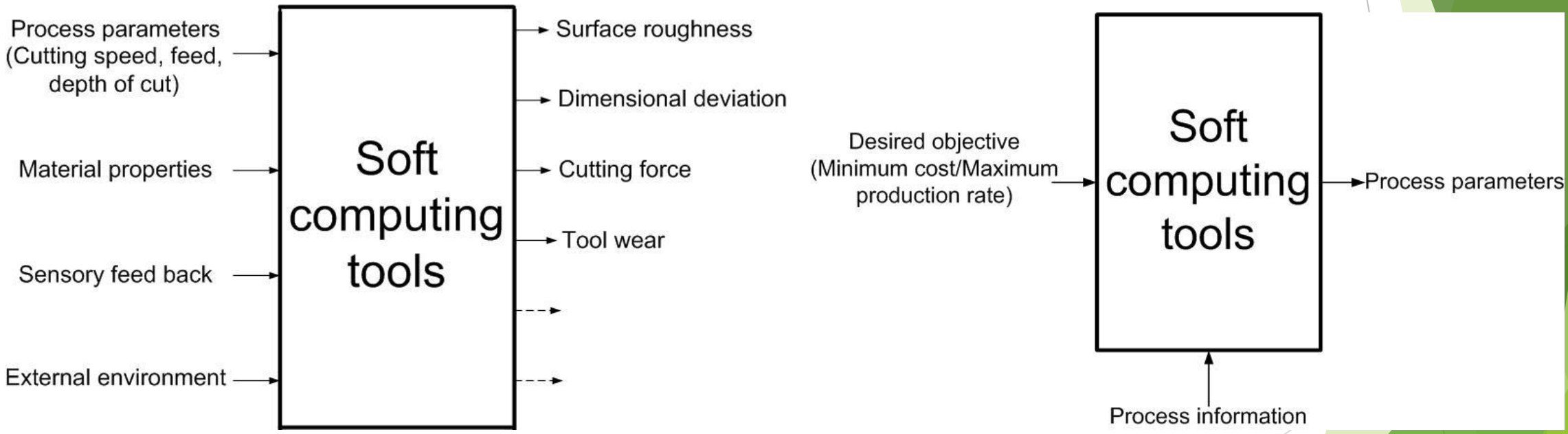
Role of cloud computing





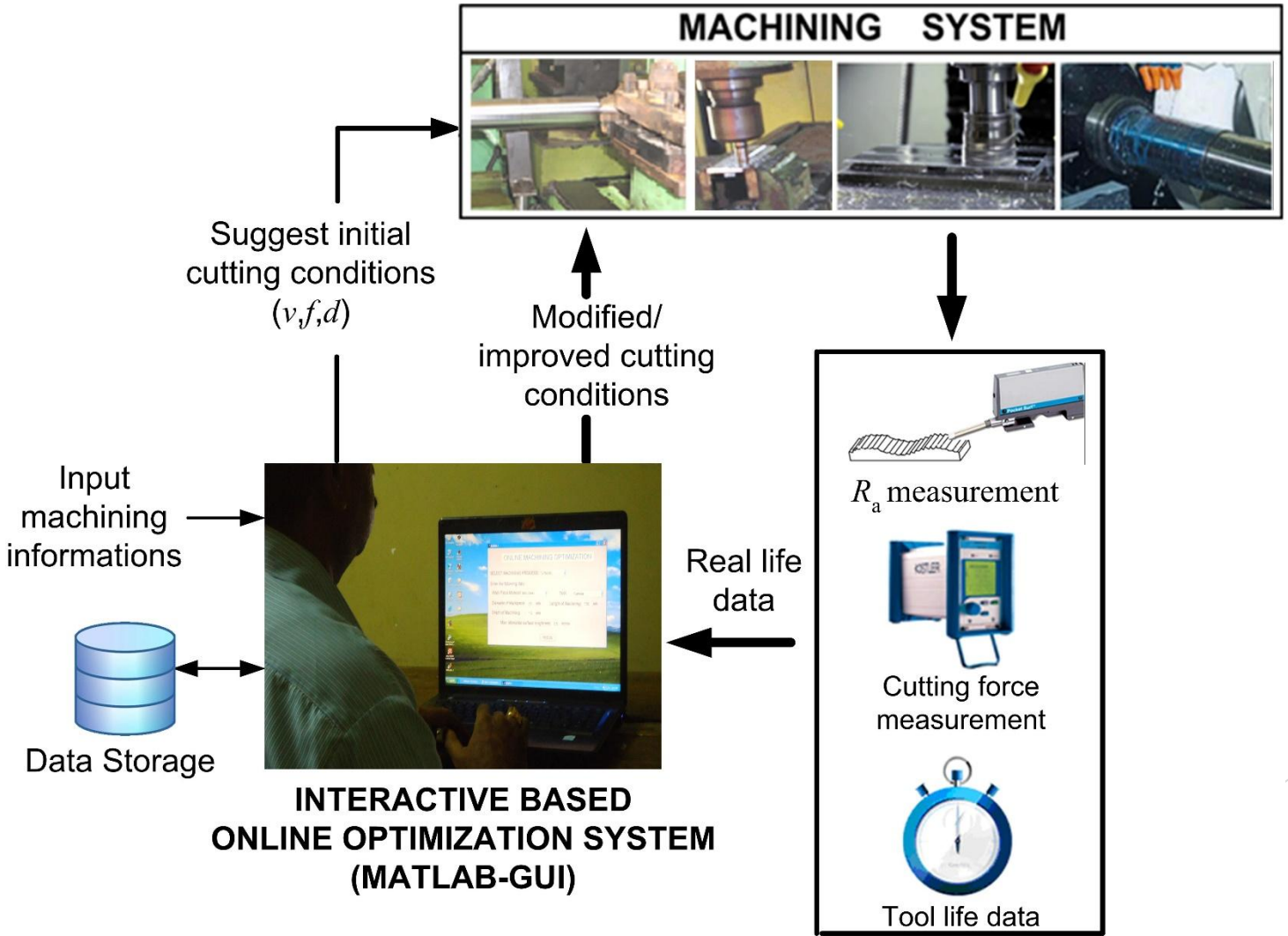
M. Chandrasekaran, 2014, **Cloud Computing Based Machining Optimization, Ph.D. Thesis, IIT Guwahtai**

Application of soft computing

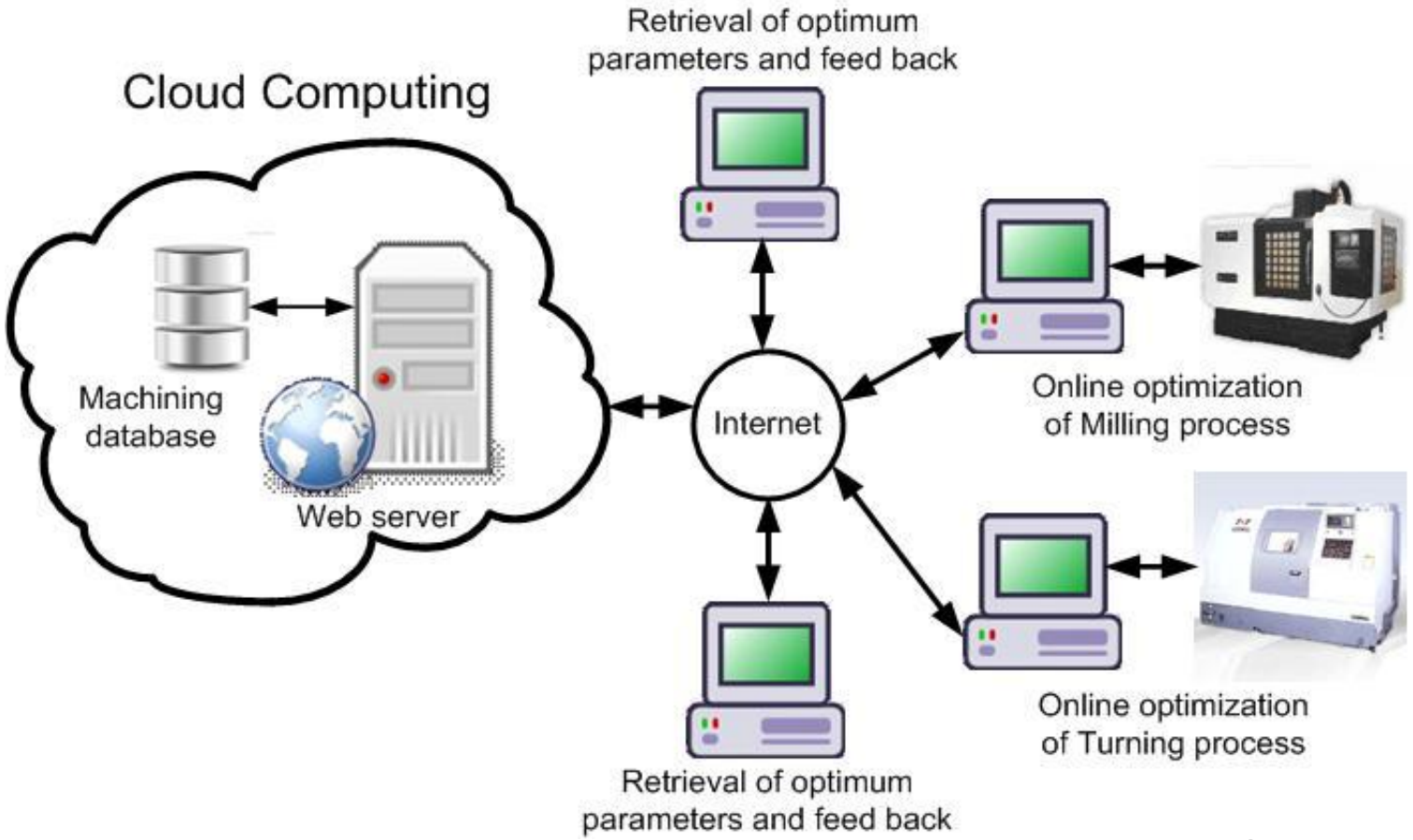


Chandrasekaran, M., Muralidhar, M., Krishna, C.M. *et al*. Application of soft computing techniques in machining performance prediction and optimization: a literature review. *Int J Adv Manuf Technol* **46,** 445–464 (2010). https://doi.org/10.1007/s00170-009-2104-x

Online optimization of machining



Cloud Computing based optimization



The National Institute of Standards and Technology defines cloud computing as follows:

"Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (*e.g.*, networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management efforts or service provider interactions."

History of Cloud Computing

- In the year 1961, John McCarthy first proposed the idea of computation being delivered as a public utility.
- He suggested in a speech at MIT (Massachusetts Institute of Technology) that computing can be sold like a utility, like electricity or water.
- In the year 1999, Salesforce.com started delivering applications to users through website and this way the dream of computing sold as utility started in reality.
- In 2002 Amazon started Amazon Web Services, providing services like storage, computation and even human intelligence.
- The turning point in the evolution of cloud computing came with the arrival of browser based cloud enterprise applications. Google Apps, Microsoft Windows Azure, and companies like Oracle and HP have joined in the main stream of cloud computing.

Cloud computing providers offer services that can be mainly grouped into three categories: infrastructure as a service (*IaaS*), software as a service (*SaaS*), and platform as a service (*PaaS*).

- Infrastructure as a service (*IaaS*): In this model, the infrastructure facilities such as storage and computing capabilities (high performance servers) are provided as a service. These services are offered on a pay-as-you-go/use basis. The consumers of *IaaS* are usually engineers and mangers, who need access to these computing resources.

- Software as a service (*SaaS*): In this model, the application and software are offered as a service. The application running on the cloud can be accessed by the user on payable basis. The user does not require installing and running the application on the computer. In the traditional system user buys licensed software and gains ownership for its maintenance and installation whereas *SaaS* is cost effective and gives significant advantage to the user. The examples are the applications such as e-mail, Facebook, and Twitter as people use everyday.

- Platform as a service (*PaaS*): In this model the platform is provided as a service. This includes the cloud infrastructure such as network, servers and operating systems. These are used for the development and deployment of applications. The main difference between *SaaS* and *PaaS* is that *SaaS* only hosts completed cloud applications whereas *PaaS* offers a development platform that hosts both completed and in-progress cloud applications. This requires *PaaS* to possess development infrastructure including programming environment, tools, and configuration management in addition to supporting application hosting environment. Microsoft windows Azure Platform Services, Google Applications, and Amazon's Relational Database Services are some of examples of *PaaS*.

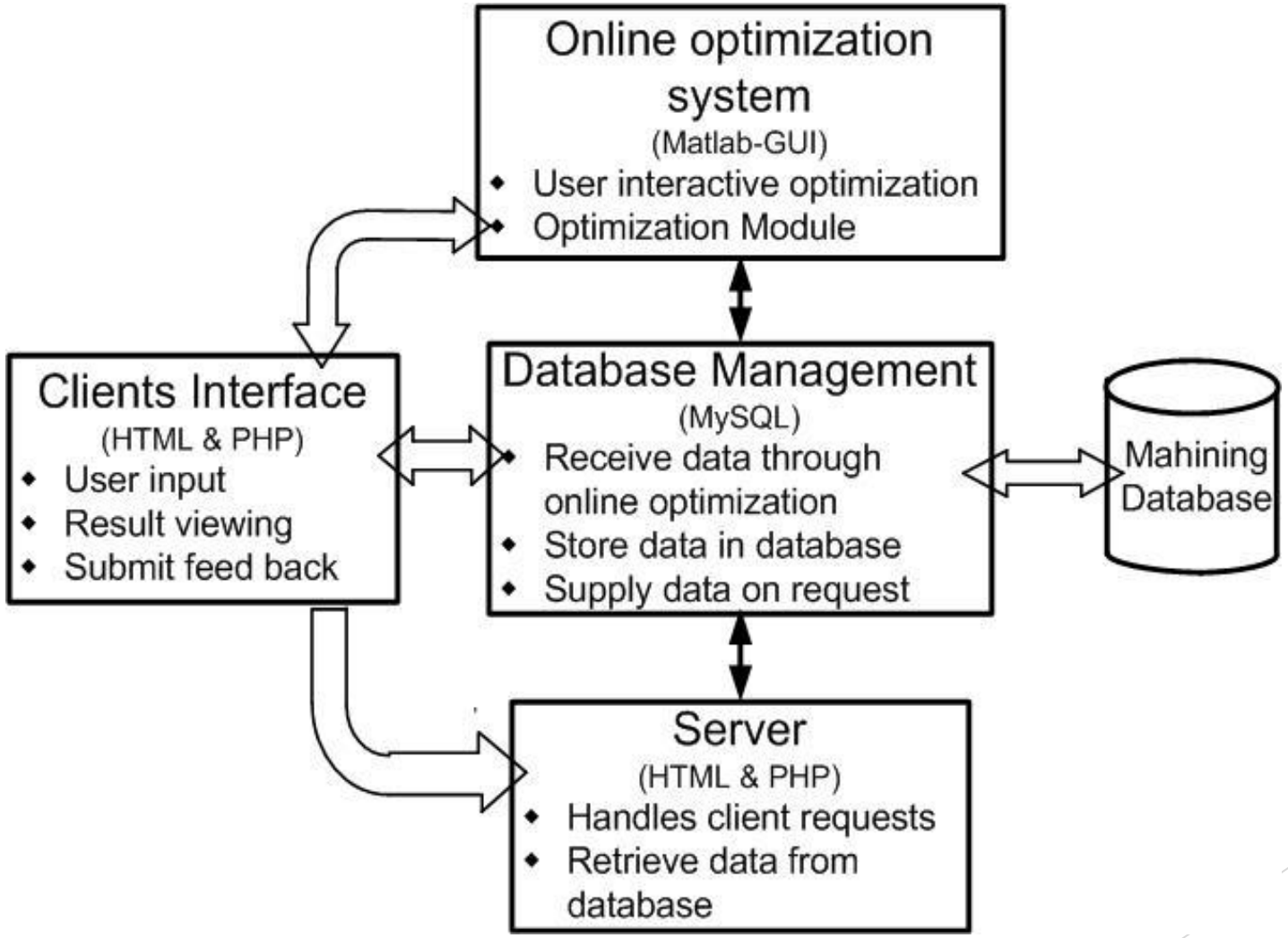
Data storage as a Service (DaaS), Design as a Service (DaaS), Manufacturing as a Service (MFGaaS), Experimentation as a Service (EaaS),

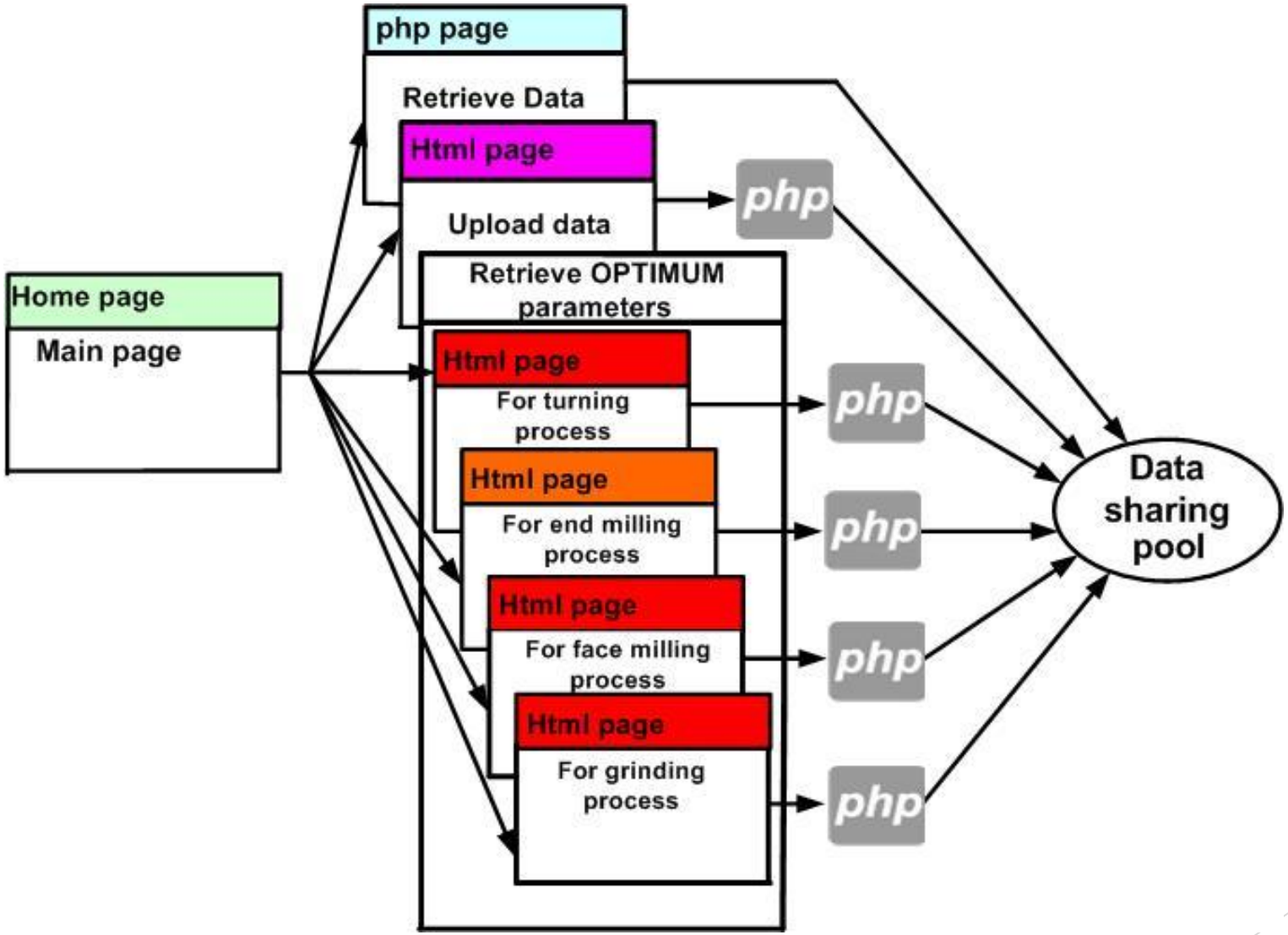
- **Private cloud: “The cloud infrastructure is operated solely for an organization. It may be operated by the organization or a third party and may exist on premise or off premise.”**
- **Community cloud: “The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns. It may be managed by the organizations or a third party and may exist on premise or off premise.”**
- **Public cloud. “The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.”**
- **Hybrid cloud. “The cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability.”**

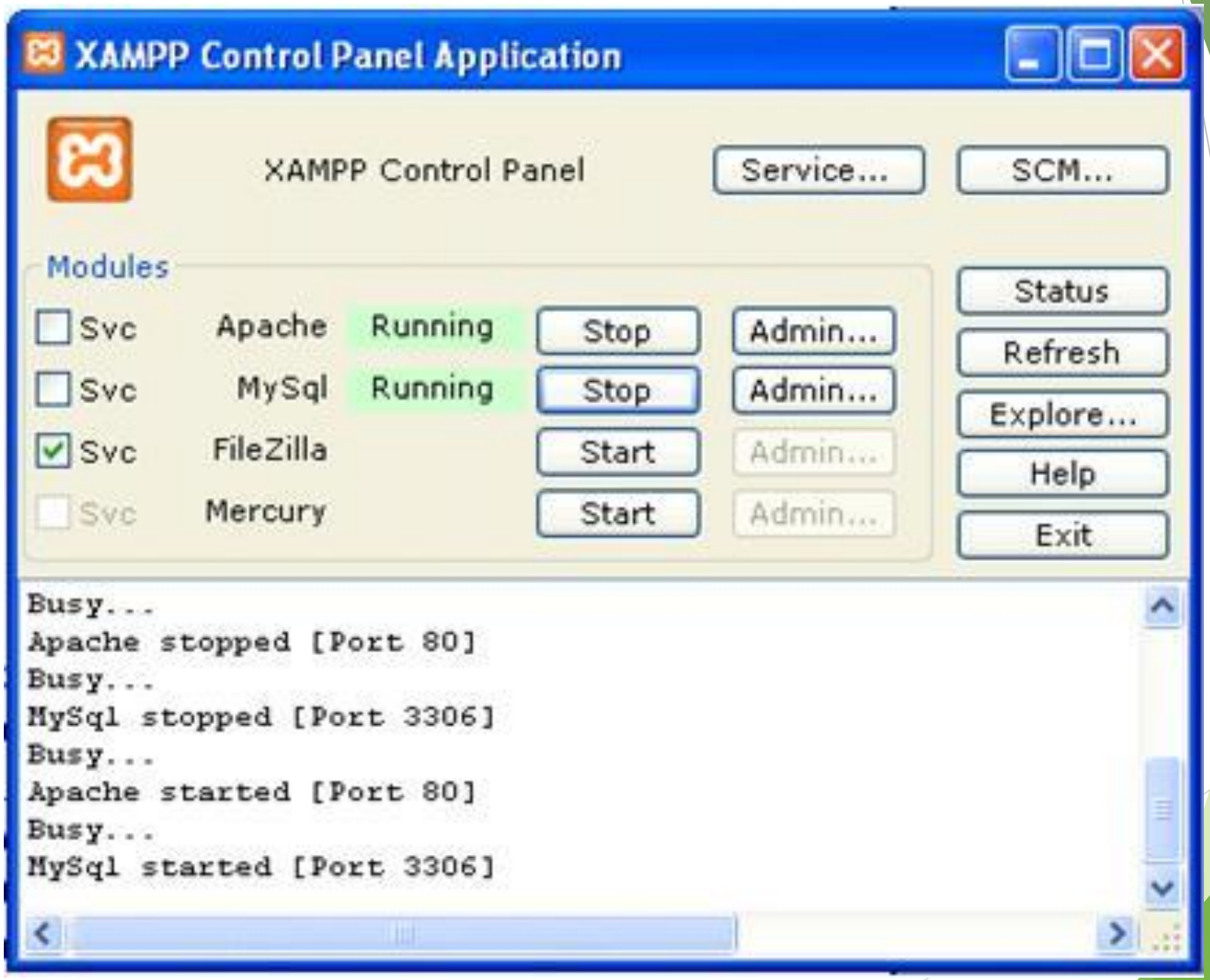
Cloud security

(i) adding authentication layers,
(ii) streamlining logging and monitoring,
(iii) identifying/selecting right application for the public cloud and
(iv) evaluating the security measure time to time.

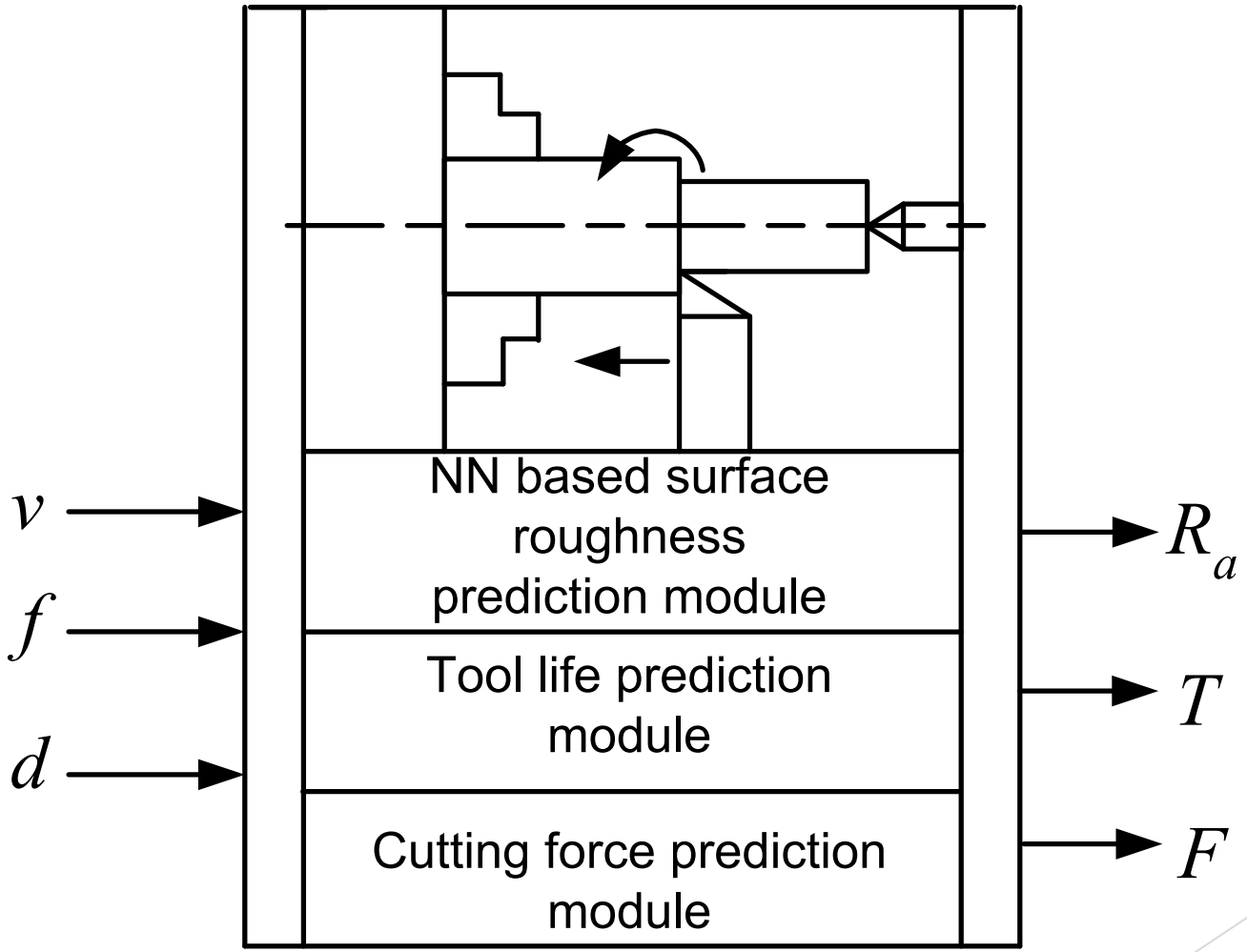
Communication structure



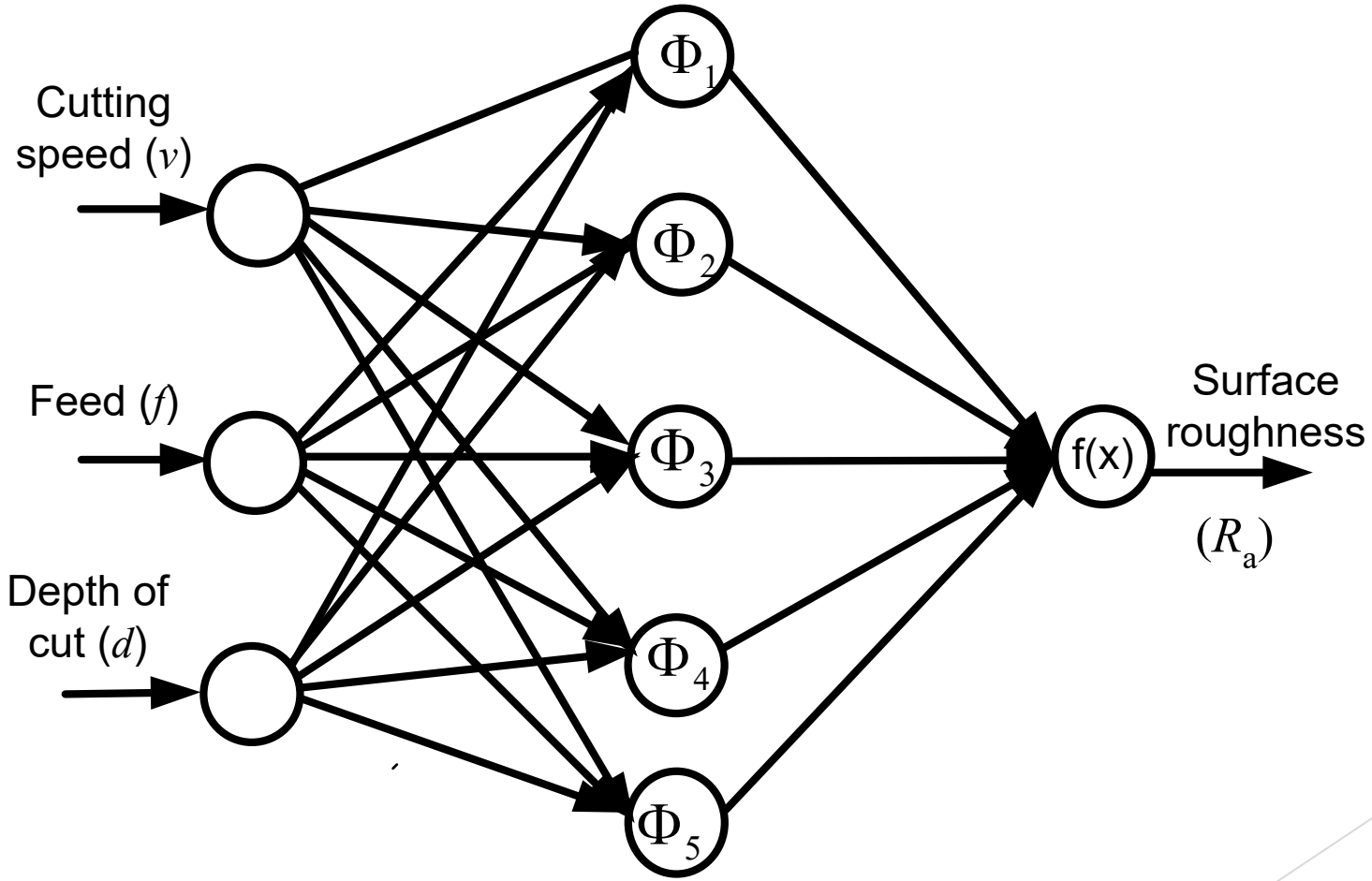

php page
Retrieve Data
Html page
Upload data
php
Retrieve OPTIMUM parameters
Home page
Main page
Html page
For turning process
php
Html page
For end milling process
php
Html page
For face milling process
php
Html page
For grinding process
php
Data sharing pool


XAMPP Control Panel Application
XAMPP Control Panel
Service...
SCM...
Modules
Svc
Apache
Running
Stop
Admin...
Svc
MySql
Running
Stop
Admin...
Svc
FileZilla
Start
Admin...
Svc
Mercury
Start
Admin...
Status
Refresh
Explore...
Help
Exit
Busy...
Apache stopped [Port 80]
Busy...
MySql stopped [Port 3306]
Busy...
Apache started [Port 80]
Busy...
MySql started [Port 3306]

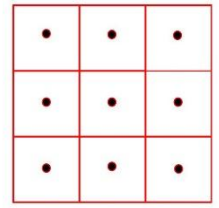
Semi-virtual lathe

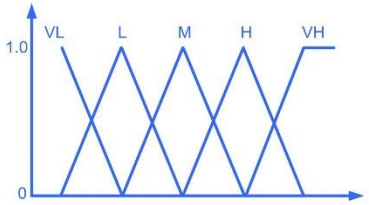


Typical Neural Network architecture

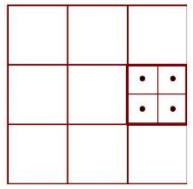


Fuzzy set based optimization





AND		y		
		L	M	H
x	L	VH	M	VH
	M	VL	L	VH
	H	VH	VH	VH



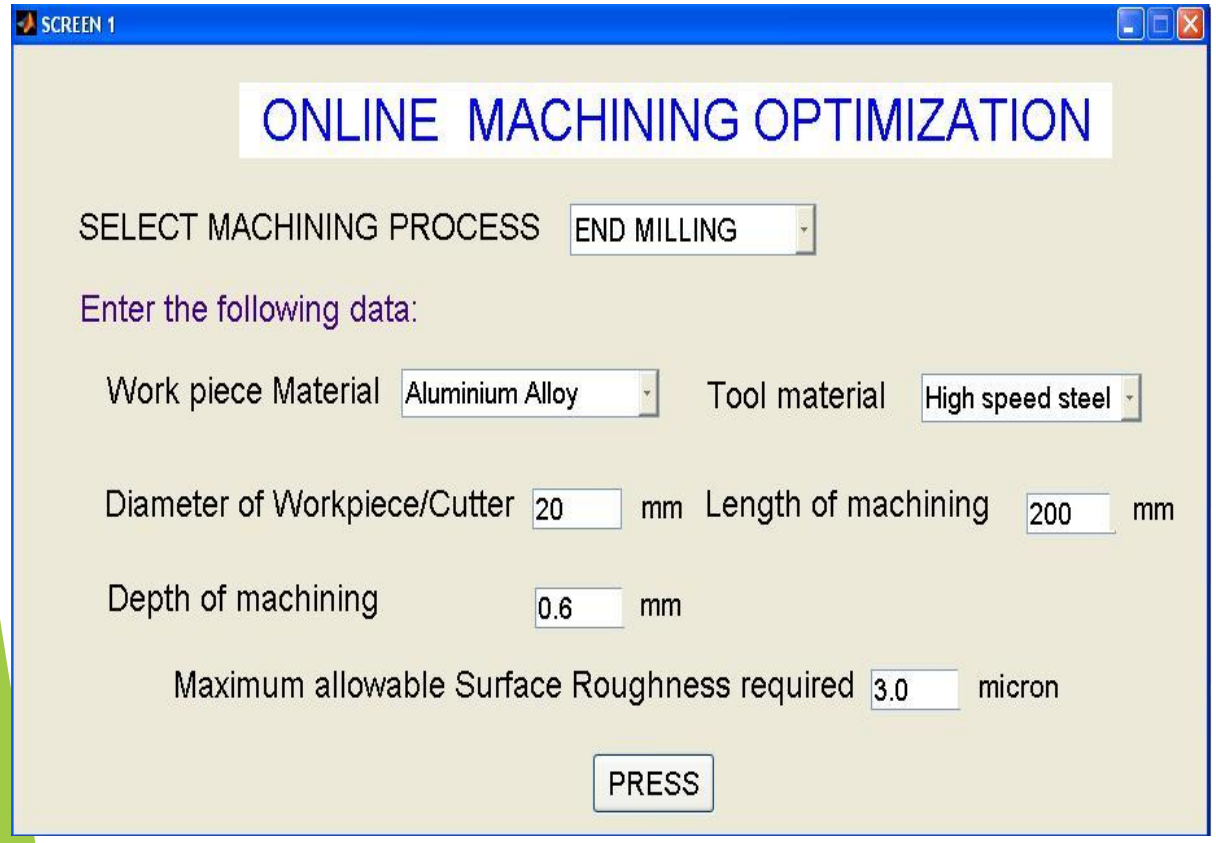
1. Divide search domain into number of cells

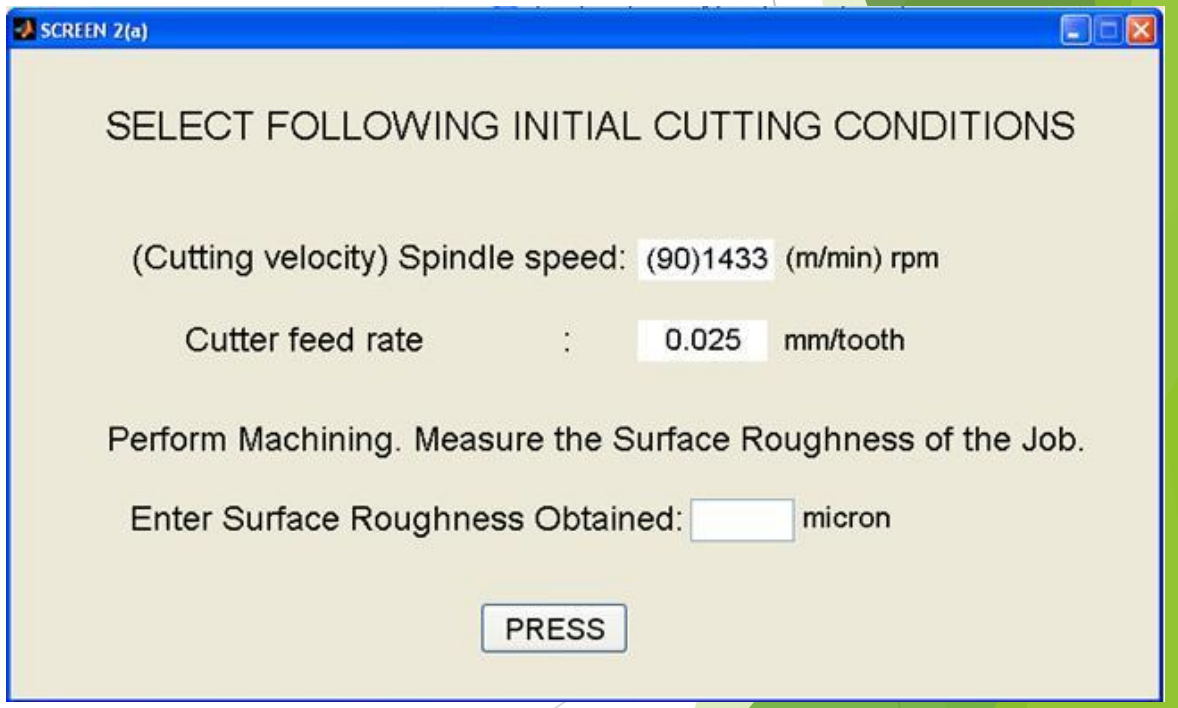
2. Fuzzify the decision variables in fuzzy subsets

3. Develop the rule base

4. Identify the optimum cell. Refine the search till desired accuracy obtained

Human-computer interface





SCREEN 7

REPORT GENERATION (ROUGH MACHINING)

THE OBTAINED ONLINE MACHINING DATA SETS ARE:
vR (m/min), dR(mm), fR (mm/rev), m, Fact (N), CR($)]

(a) INITIAL MACHINING:

140	1.0	0.04	2	386.3	2.41
154	1.0	0.04	2	386.3	2.28
154	1.0	0.05	2	408.3	1.98

(b) ITERATIVE DATA SETS:

173	1.0	0.06	2	427.2	1.69
154	1.0	0.08	2	458.8	1.53
142	1.0	0.14	2	527.1	1.24

(c) OPTIMUM CUTTING CONDITION:

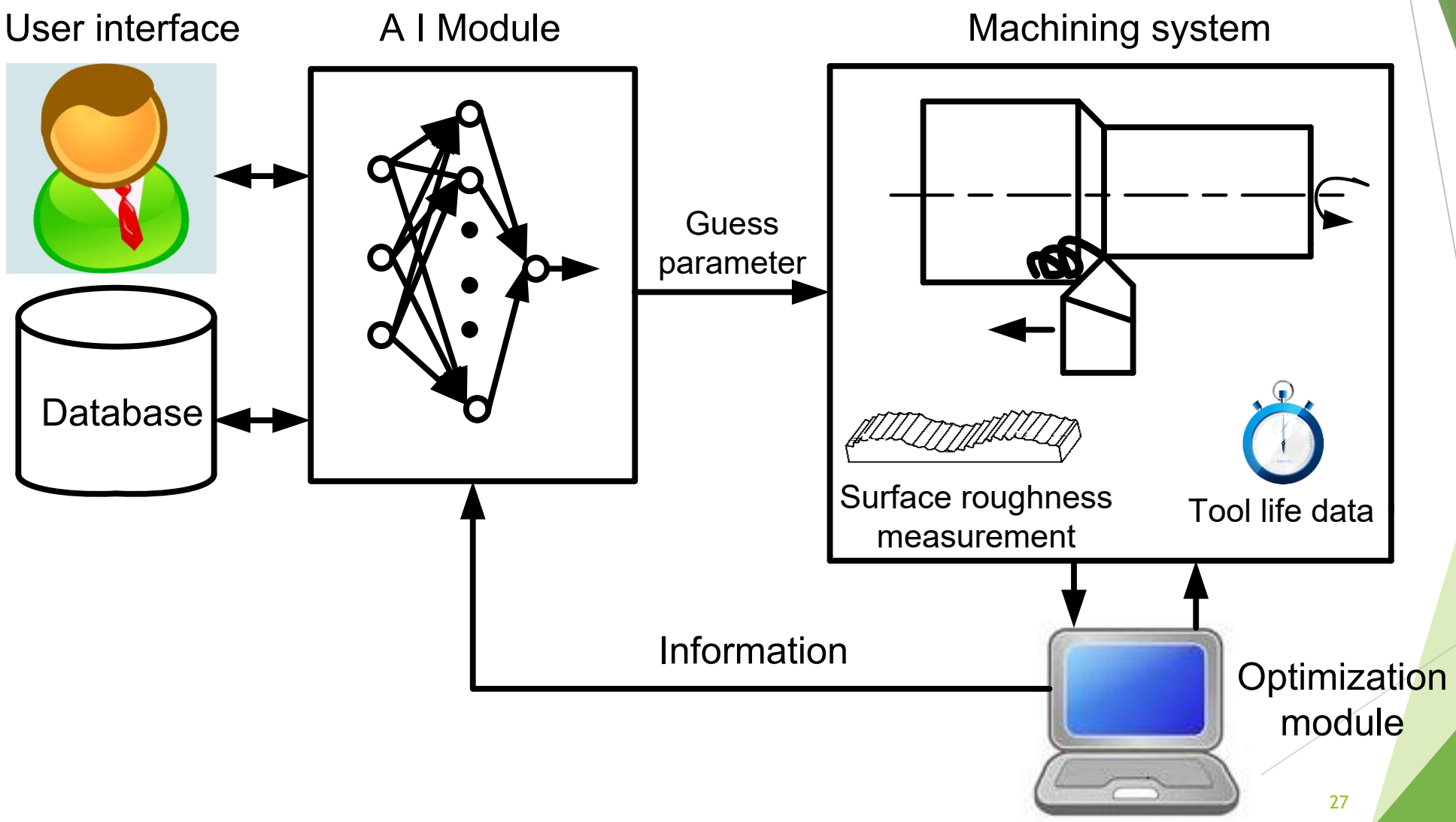
142	1.0	0.14	2	527.1	1.24

(d) ROUGHING COST / JOB: $ 1.24

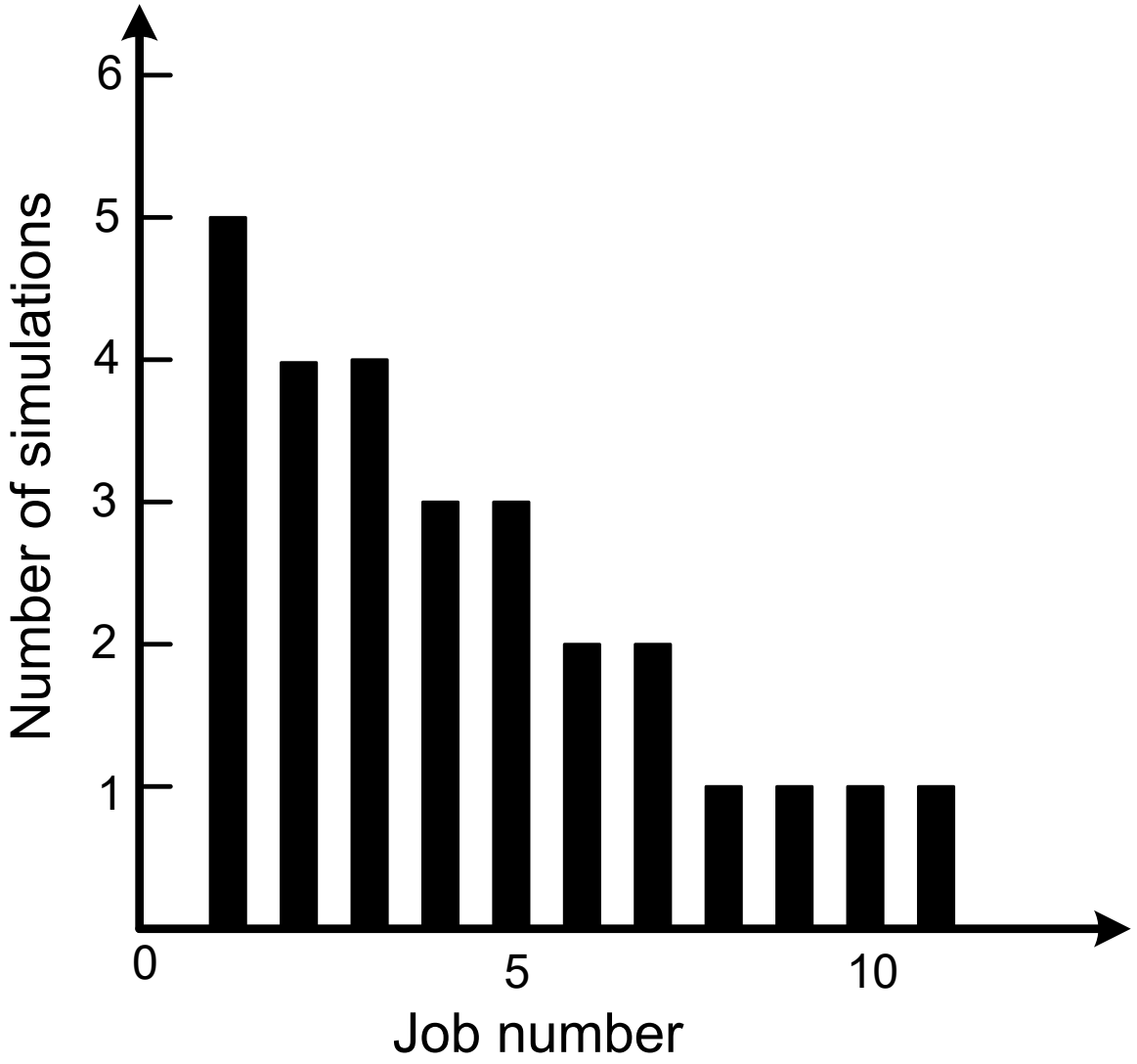
BACK TO MAIN

FINISH OPERATION

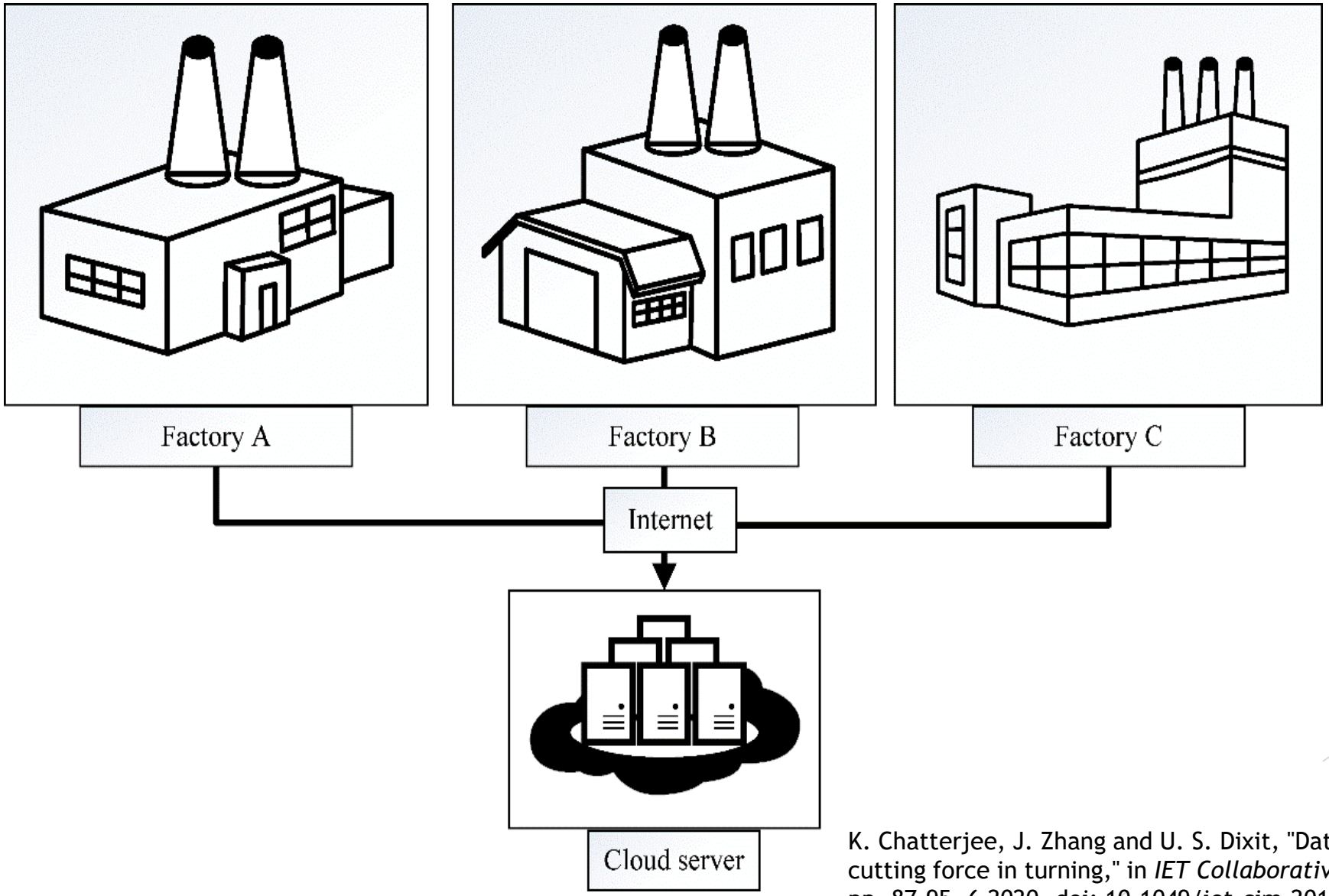
Online learning



Reduction in simulation runs with learning

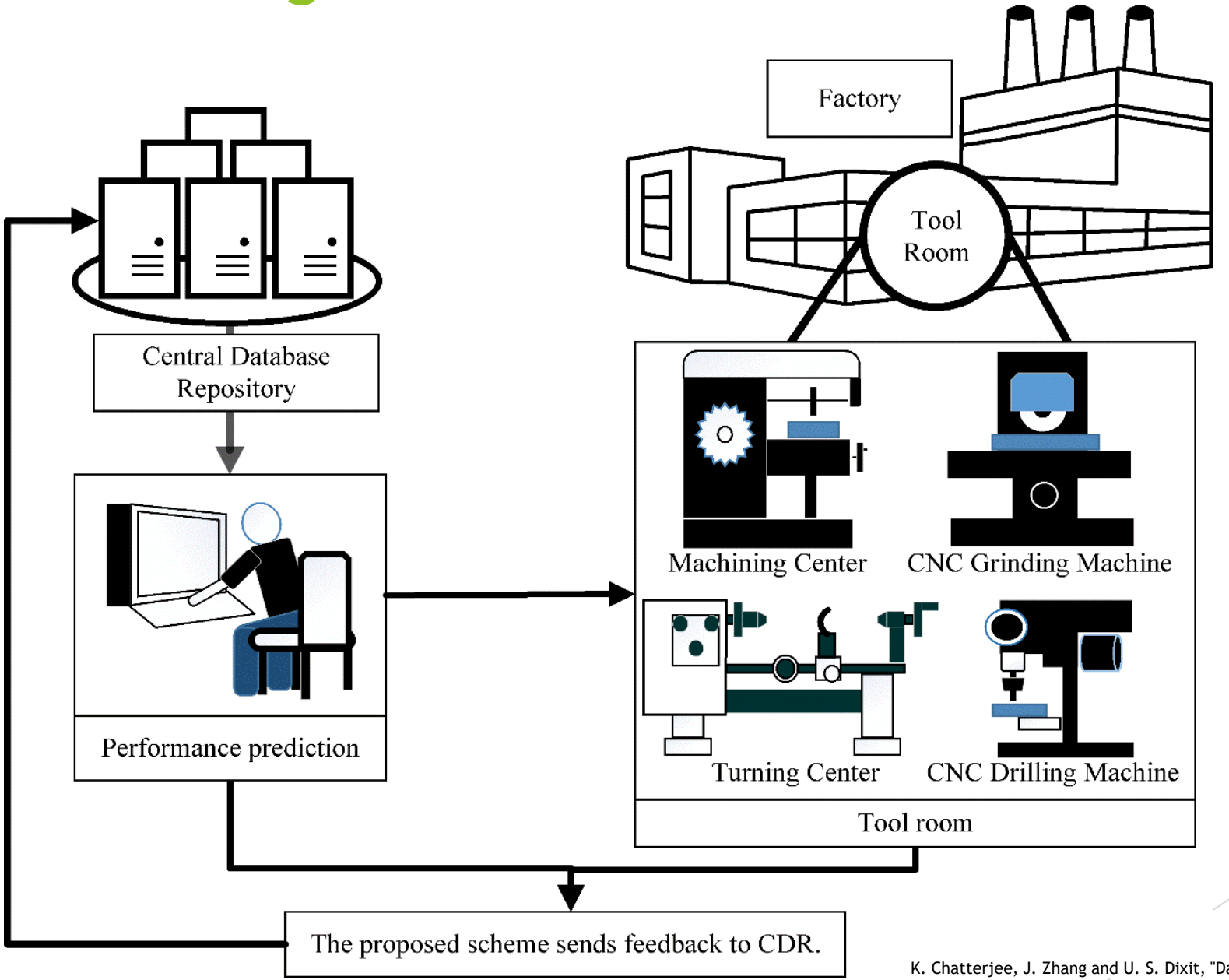


Application of big data in manufacturing



K. Chatterjee, J. Zhang and U. S. Dixit, "Data-driven framework for the prediction of cutting force in turning," in *IET Collaborative Intelligent Manufacturing*, vol. 2, no. 2, pp. 87-95, 6 2020, doi: 10.1049/iet-cim.2019.0055.

Data Management and Use



K. Chatterjee, J. Zhang and U. S. Dixit, "Data-driven framework for the prediction of cutting force in turning," in *IET Collaborative Intelligent Manufacturing*, vol. 2, no. 2, pp. 87-95, 6 2020, doi: 10.1049/iet-cim.2019.0055.

Challenging issues in Big Data Applications

- Handling of outliers
- Data security
- Avoiding data explosion
- Dealing with scarcity of data
- Assessing veracity
- Efficient way of prediction (Piecewise linear prediction)

How can digital manufacturing be a reality?

- Modelling and simulation is an important part.
- Physics of the process should be understood clearly.
- CBDM (Cloud Based Design and Manufacturing)
- Cloud based design uses cloud computing, Service-oriented architecture, Web 2.0 and semantic web technologies.

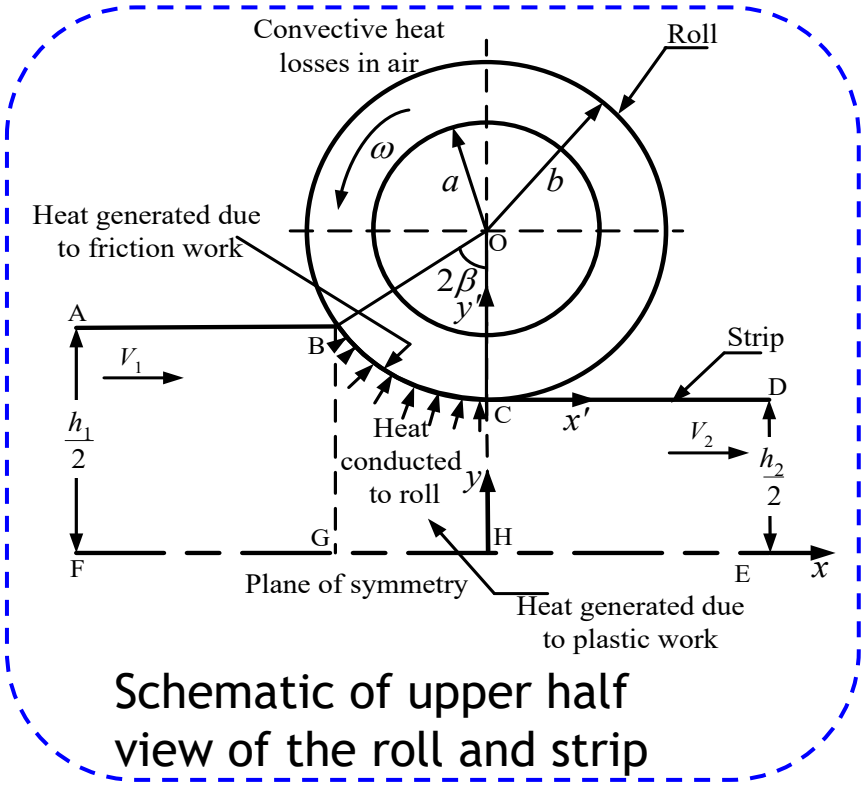
Purpose of modeling of a Manufacturing Process

- **The power requirement**
- **Pressure in the die and tools**
- **Stresses, strains and strain rate distributions in the material during processing**
- **Residual stress in the product**
- **Defects**
- **Geometric accuracy**
- **Surface integrity**
- **Mechanical and metallurgical properties**
- **Microstructure**

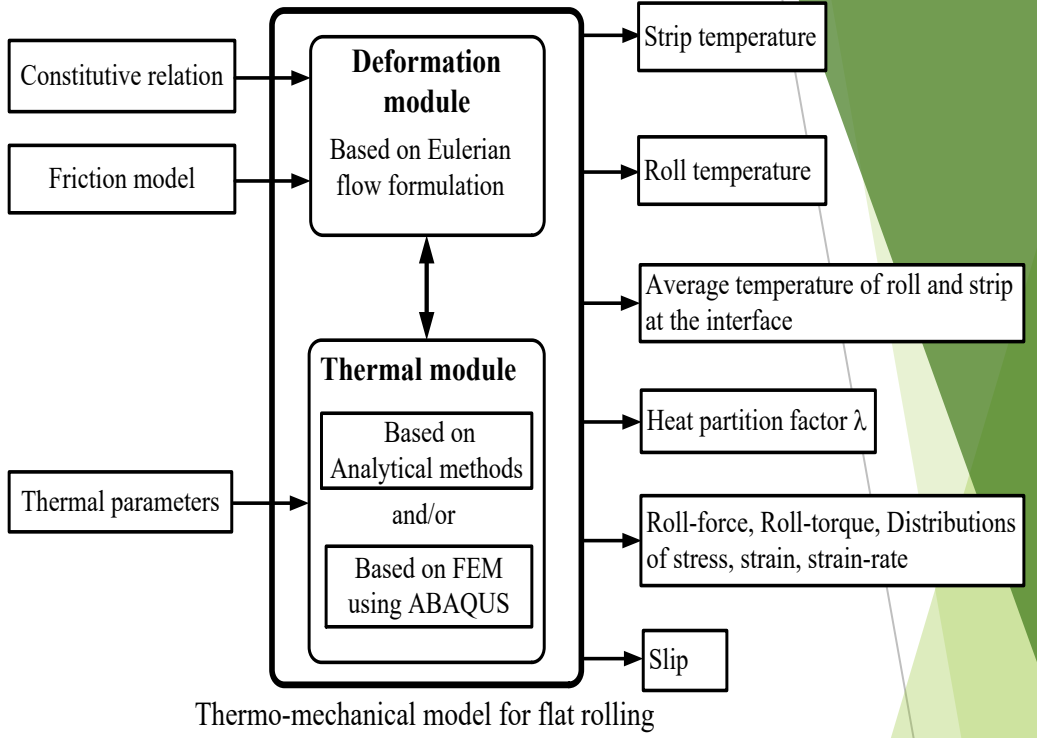
Modelling methods

- Slab Method
- Upper bound
- Lower Bound
- Slip Line
- Finite Element Method
- Finite Difference Method
- Finite Volume Method
- Boundary Element Method
- Molecular Dynamics Simulation
- Crystal Plasticity

Thermal Modelling: warm flat rolling

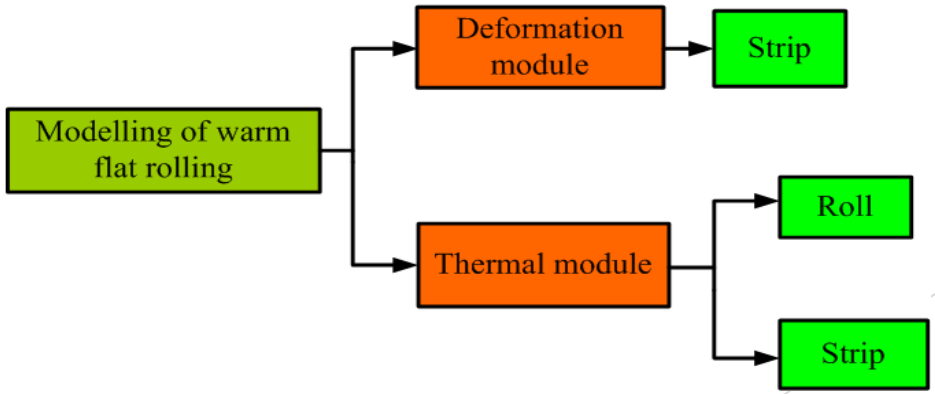


Schematic of upper half view of the roll and strip

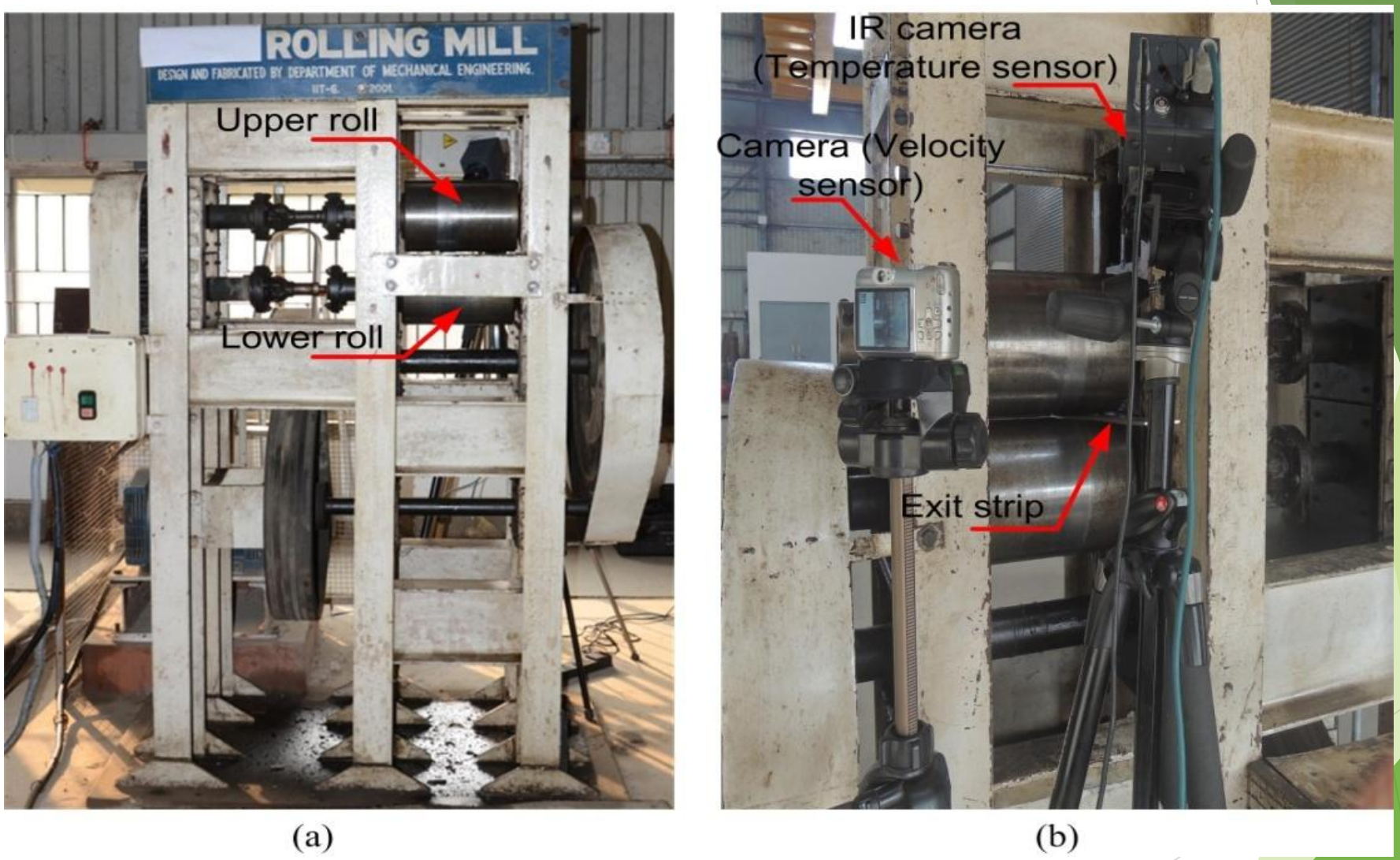


Thermo-mechanical model for flat rolling

An overview of thermo-mechanical model of plane strain rolling



Inverse Modelling for estimation of parameters

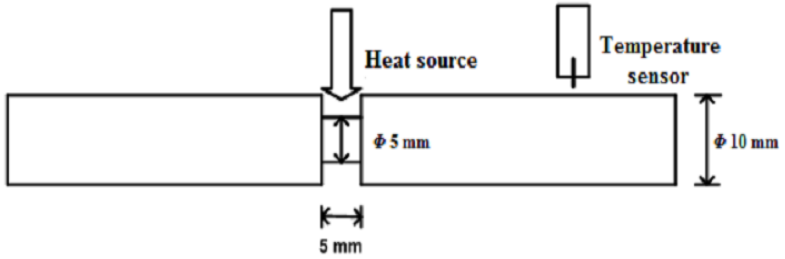


A laboratory rolling mill (a) front view and (b) arrangement for measuring the temperature and velocity of exit strip at the rear side

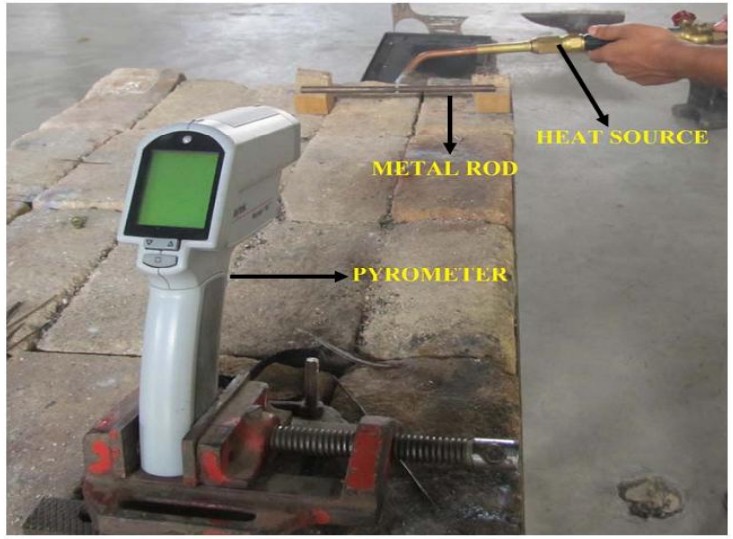
Inverse determination of heat flux and thermal properties



(a)



(b)



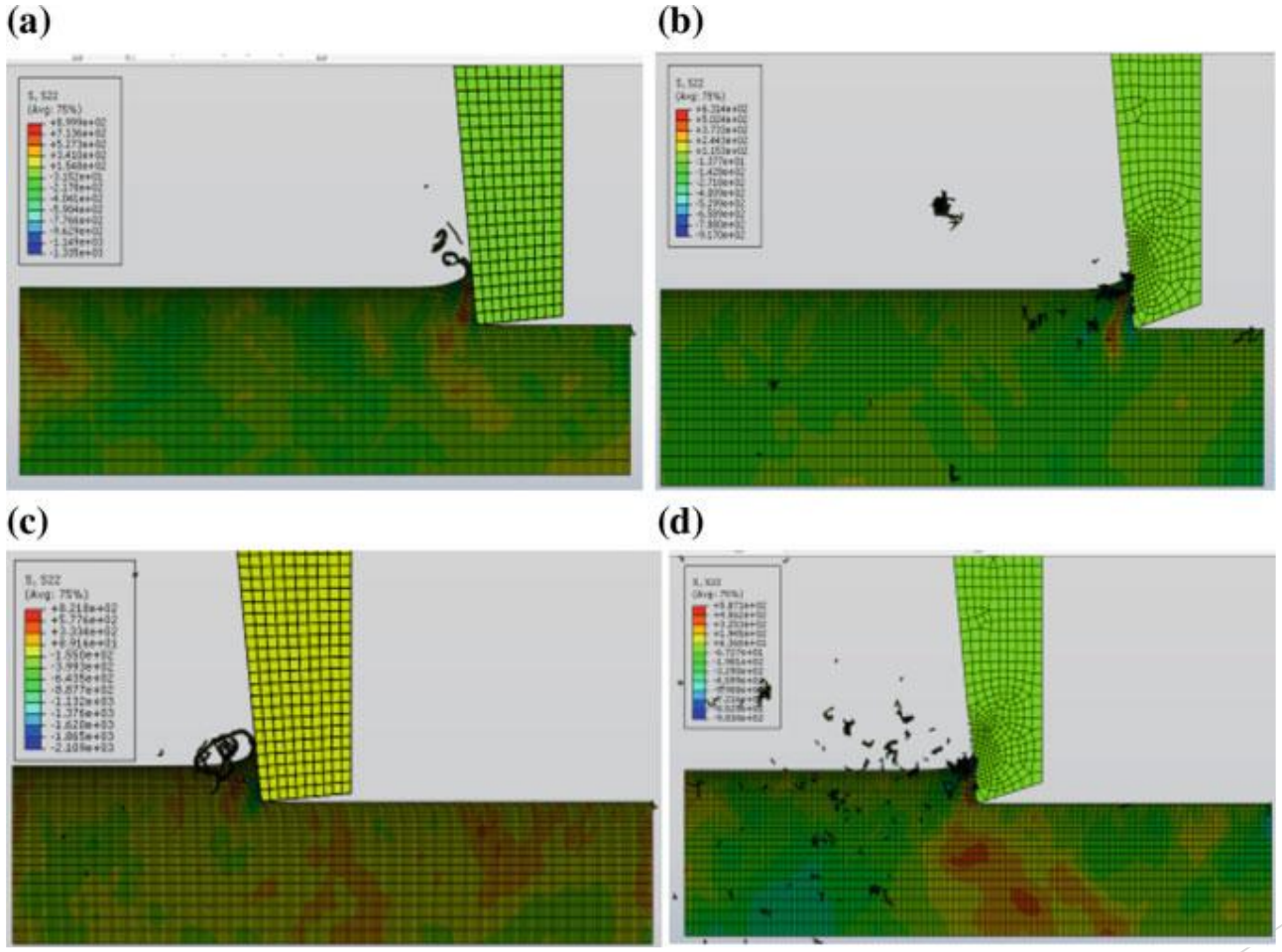
(c)



(d)

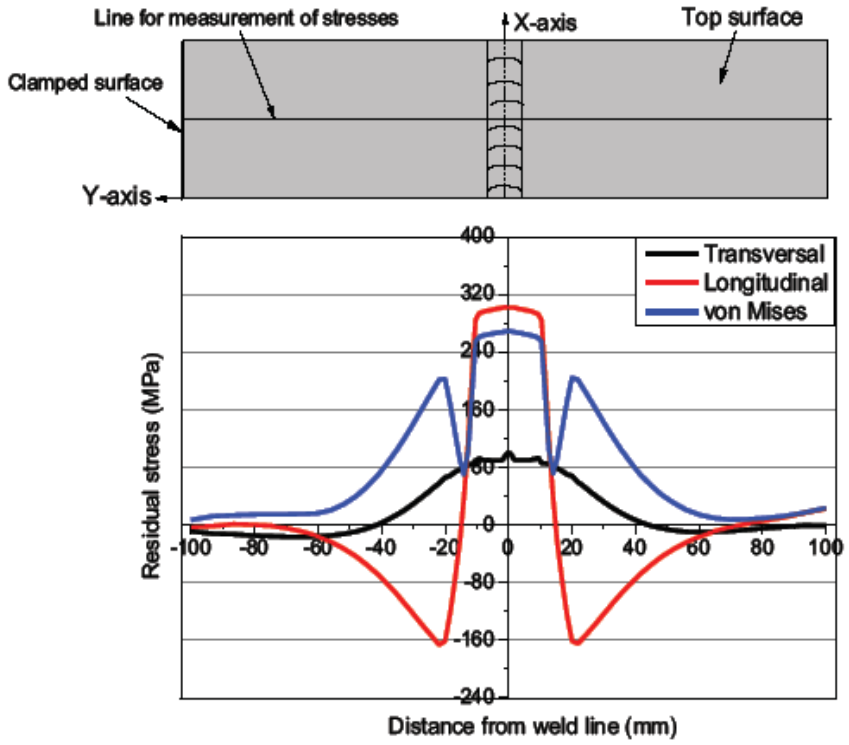
(a) Applied heat source (b) schematic (c) arrangement of temperature measurement (d) laser based pyrometer

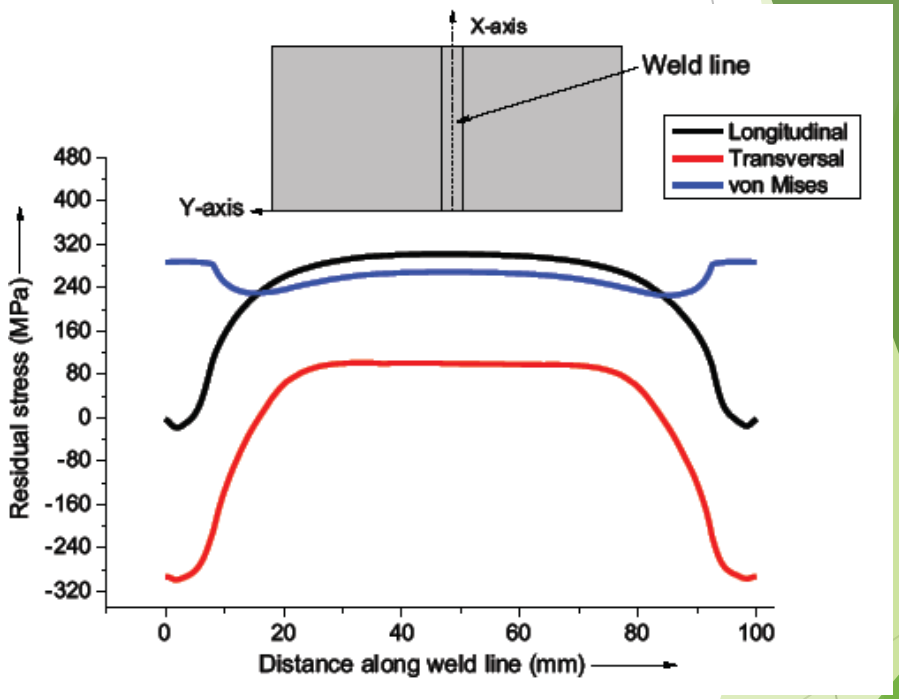
Residual Stresses in Machining



Varun Sharma, Pulak M. Pandey, Uday S. Dixit, Anish Roy, Vadim V. Silberschmidt, 2018, Ultrasonic assisted turning: a comparative study of surface integrity, Precision Product-Process Design and Optimization: Select Papers from AIMTDR 2016, Edited by S.S. Pande and U.S. Dixit, Springer, Singapore, pp. 337-360.

Residual Stress in Welding





R. Bhadra, P. Pankaj, P. Biswas and U.S. Dixit, 2018, Thermo-mechanical analysis of CO_2 laser butt welding on AISI 304 steel thin plates, International Journal of Steel Structures, Vol. 19, Issue 1, pp. 14-27

How do we estimate residual stresses?

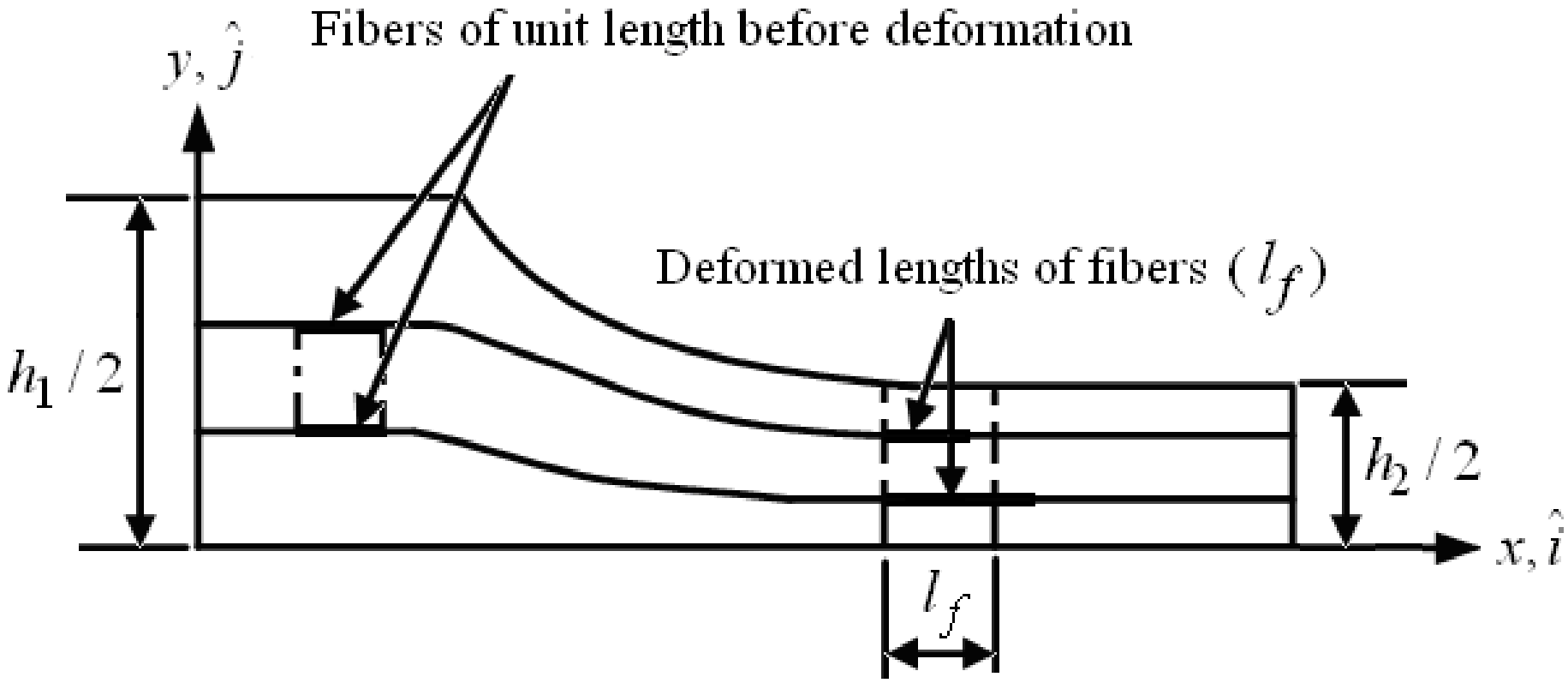
- One needs to do elastic-plastic analysis.
- FEM is a versatile method.
- Two ways: Updated Lagrangian and Eulerian.
- Updated Lagrangian methods takes a lot of computational time. Numerical errors accumulate.
- Eulerian method has not proved to be very effective in the estimation of residual stresses.

Dixit, U. S., and P. M. Dixit. "A study on residual stresses in rolling." *International Journal of Machine Tools and Manufacture*, Vol. 37, Issue 6 (1997): 837-853.

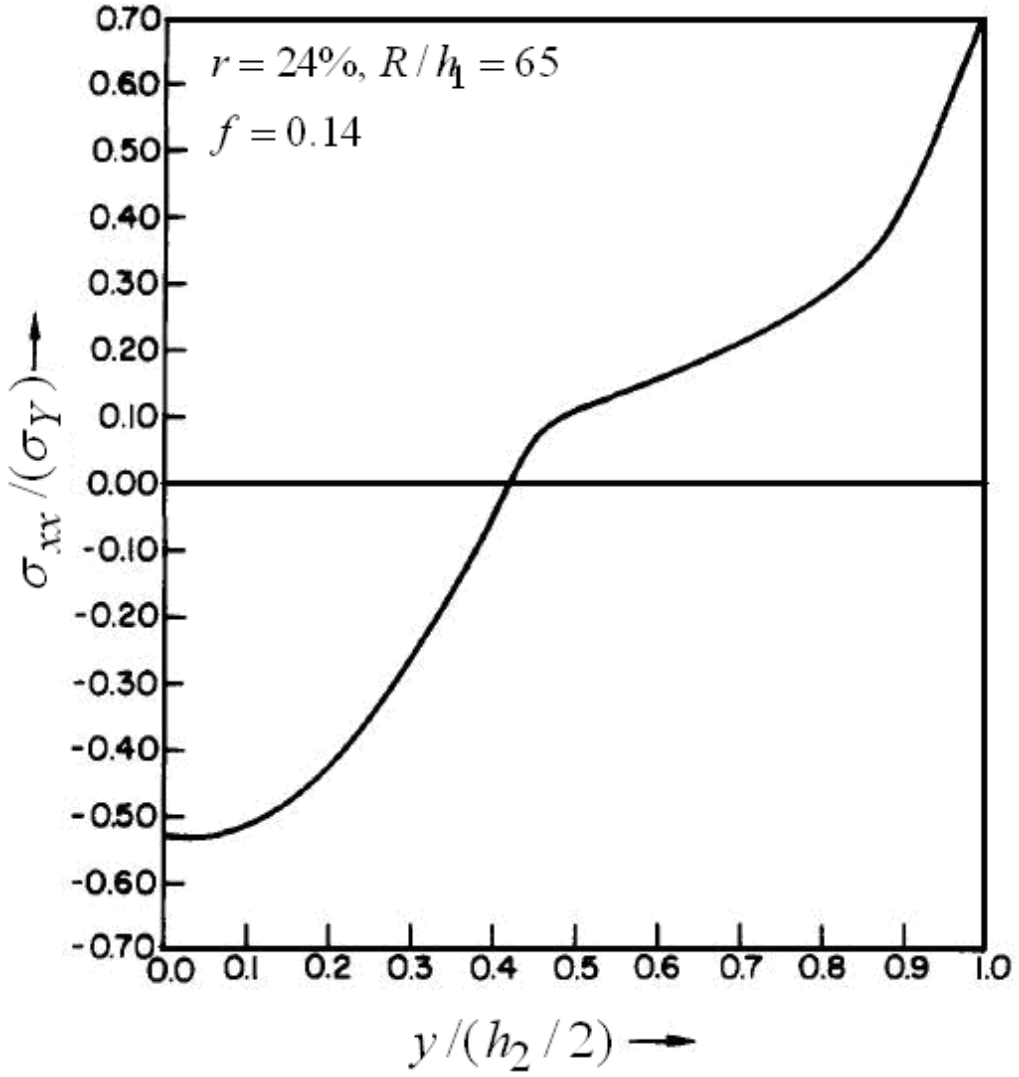
Discussed various Eulerian Methods:

- Mixed formulation
- Method of multiplicative decomposition of deformation gradient
- Rate formulation
- Simplified approach to find longitudinal residual stresses in rolling

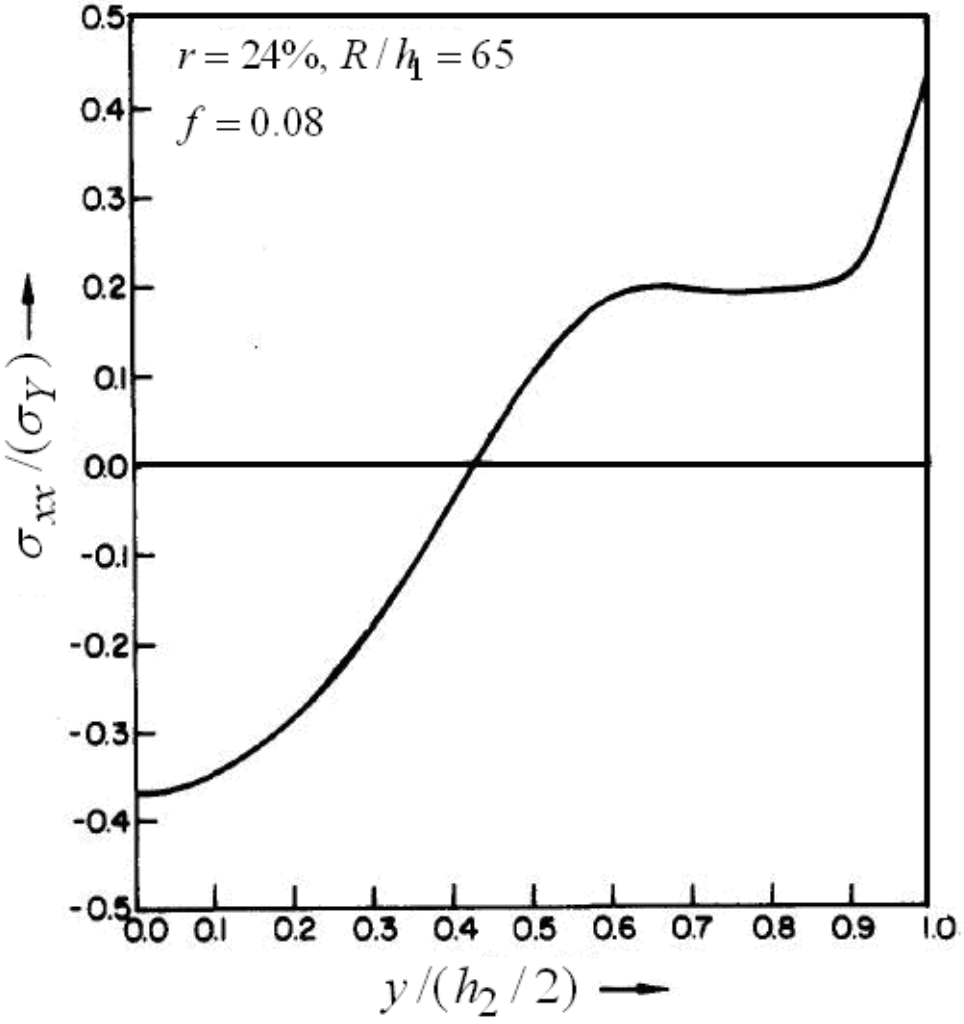
Non-uniform stretching of two fibers



Distribution of longitudinal residual stress across the thickness (high friction)



Distribution of longitudinal residual stress across the thickness for low friction



Difficulties in measurement

- Techniques are expensive
- Some relative cheaper techniques take a lot of time and require much skill
- Some techniques measure the stresses only on the surface
- Very difficult to measure online
- No robust technique to measure all types of residual stresses

Multiscale Modelling- Combining continuum mechanics with microstructure

- Many times research carry out only modelling with continuum approach disregarding the materials' aspects.
- Die shape optimization in extrusion was carried out for minimizing redundant work of deformation, but redundant work is not useless; it may improve the strength.
- During machining, a lot of changes in material take place, which need to be modelled.